VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Agile Development Process and Devops Lab – ISWE406P

Digital Assessment – 4

Faculty: Dr. Bala Ganesh N

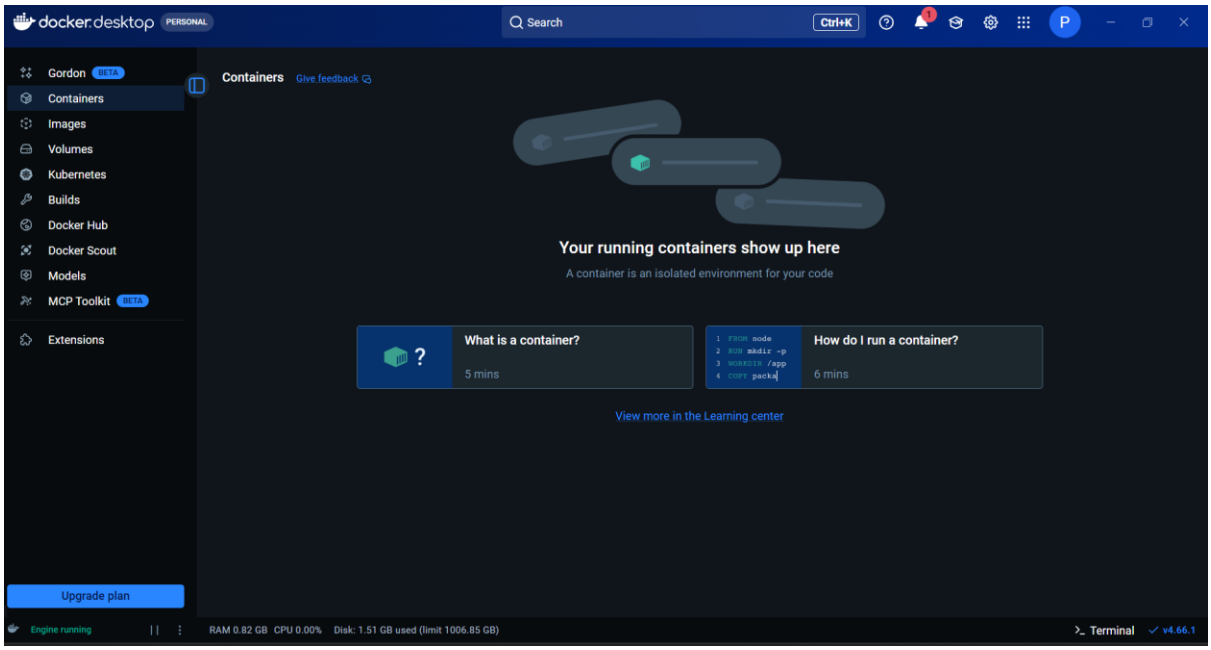
Slot: L3 + L4

Name: Pradheep.R

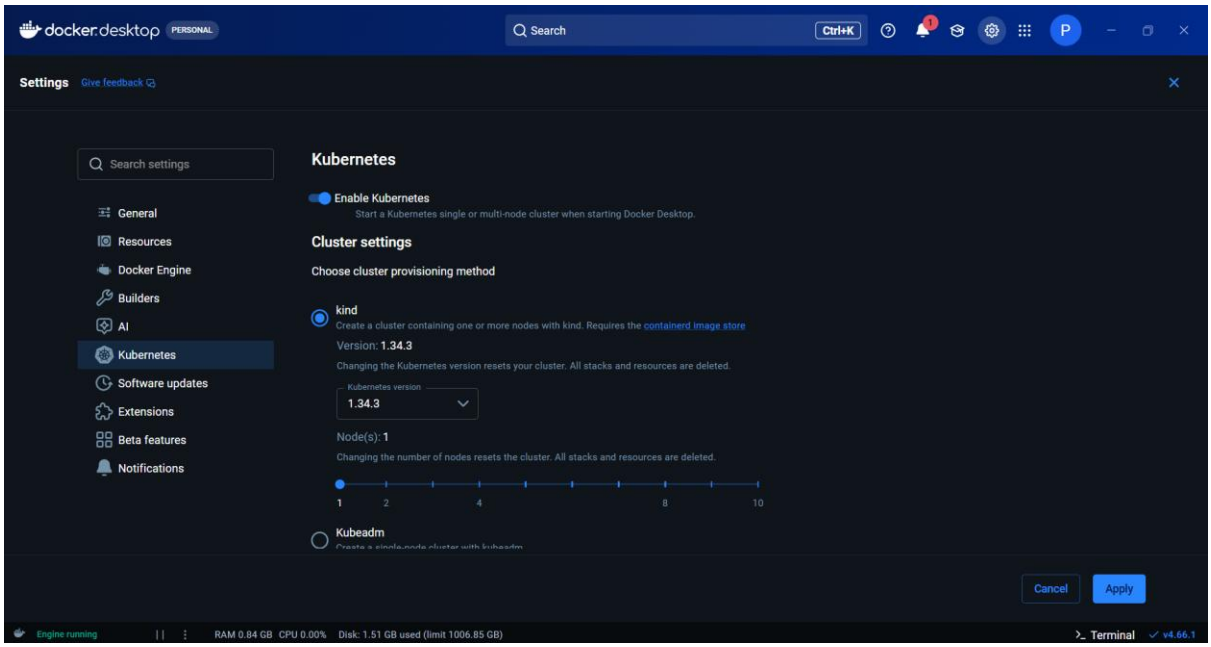
Register Number: 23MIS0518

Step 1:

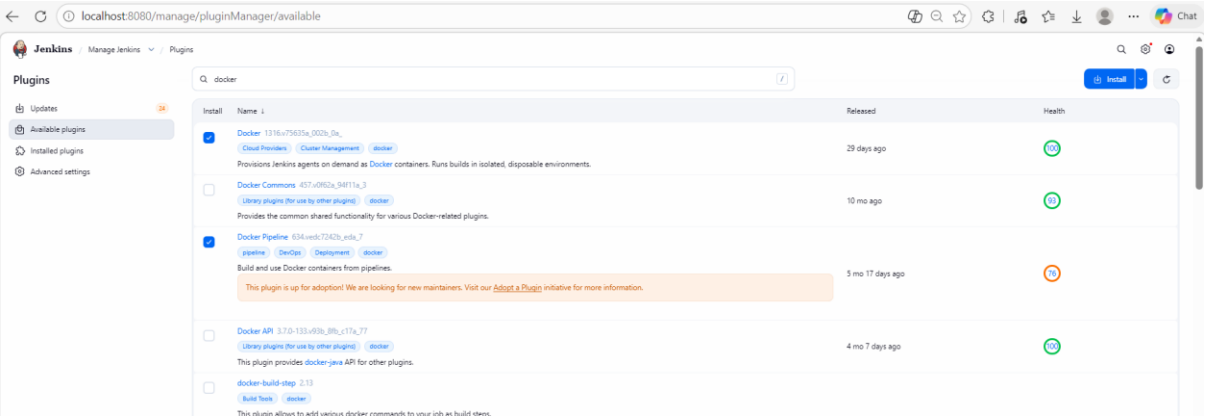
Install Docket Desktop

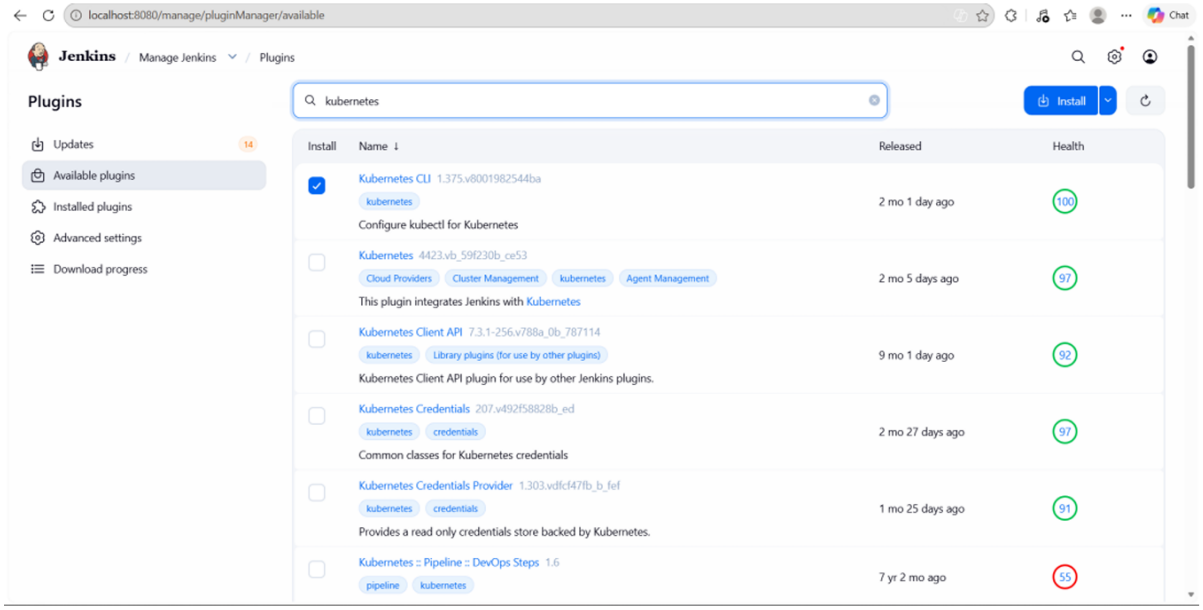


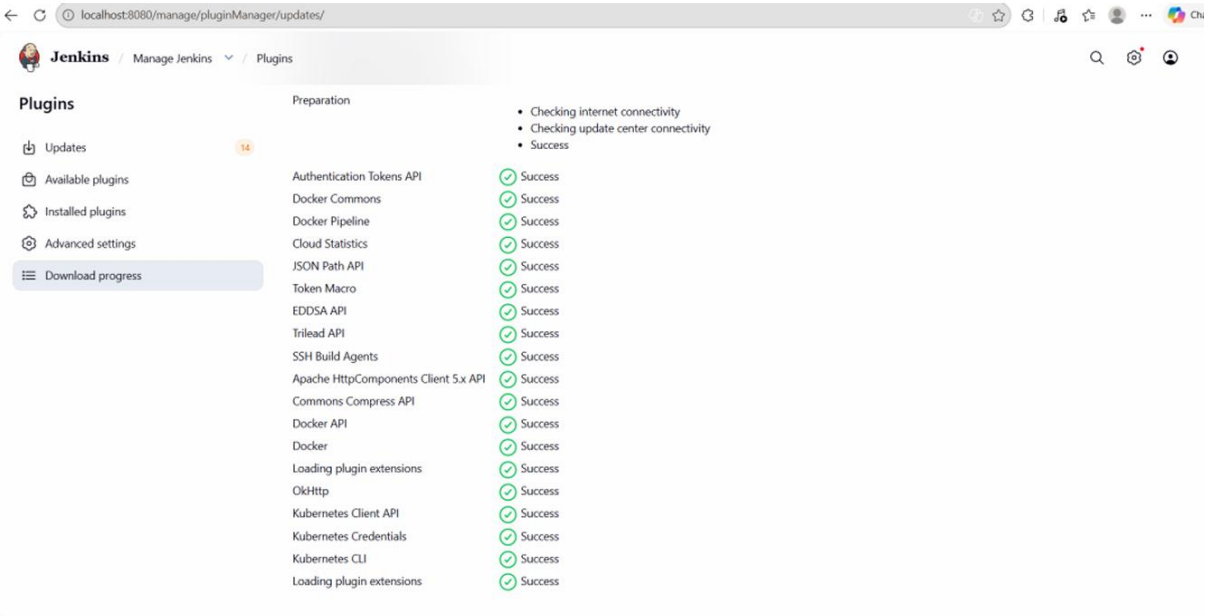
Enable Kubernets



Enable three plugins (Docker, Docker Pipeline, Kubernetes CLI) and install them:

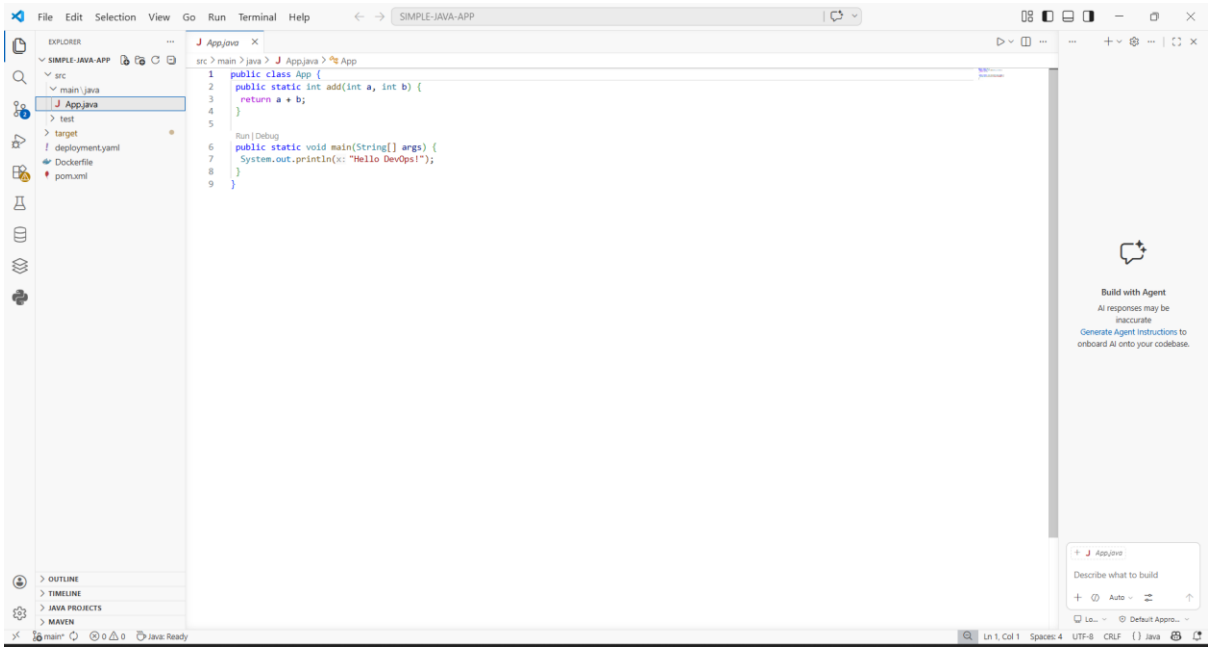

localhost:8080/manage/pluginManager/available
Jenkins / Manage Jenkins / Plugins
Plugins
docker
Install
Updates
Available plugins
Installed plugins
Advanced settings
Install
Name
Released
Health
Docker 1316.v75635a_002b_0a_
Cloud Providers Cluster Management docker
Provisions Jenkins agents on demand as Docker containers. Runs builds in isolated, disposable environments.
29 days ago
100
Docker Commons 457.v0f62a_94f11a_3
Library plugins (for use by other plugins) docker
Provides the common shared functionality for various Docker-related plugins.
10 mo ago
93
Docker Pipeline 634.vedc7242b_eda_7
pipeline DevOps Deployment docker
Build and use Docker containers from pipelines.
This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.
5 mo 17 days ago
76
Docker API 3.7.0-133.v93b_8ffb_c17a_77
Library plugins (for use by other plugins) docker
This plugin provides docker-java API for other plugins.
4 mo 7 days ago
100
docker-build-step 2.13
Build Tools docker


localhost:8080/manage/pluginManager/available
Jenkins / Manage Jenkins / Plugins
Plugins
kubernetes
Install
Updates 14
Available plugins
Installed plugins
Advanced settings
Download progress
Install
Name
Released
Health
Kubernetes CLI 1.375.v8001982544ba
kubernetes
Configure kubectl for Kubernetes
2 mo 1 day ago
100
Kubernetes 4423.vb_59f230b_ce53
Cloud Providers Cluster Management kubernetes Agent Management
This plugin integrates Jenkins with Kubernetes
2 mo 5 days ago
97
Kubernetes Client API 7.3.1-256.v788a_0b_787114
kubernetes Library plugins (for use by other plugins)
Kubernetes Client API plugin for use by other Jenkins plugins.
9 mo 1 day ago
92
Kubernetes Credentials 207.v492f58828b_ed
kubernetes credentials
Common classes for Kubernetes credentials
2 mo 27 days ago
97
Kubernetes Credentials Provider 1.303.vdfcf47fb_b_fef
kubernetes credentials
Provides a read only credentials store backed by Kubernetes.
1 mo 25 days ago
91
Kubernetes :: Pipeline :: DevOps Steps 1.6
pipeline kubernetes
7 yr 2 mo ago
55

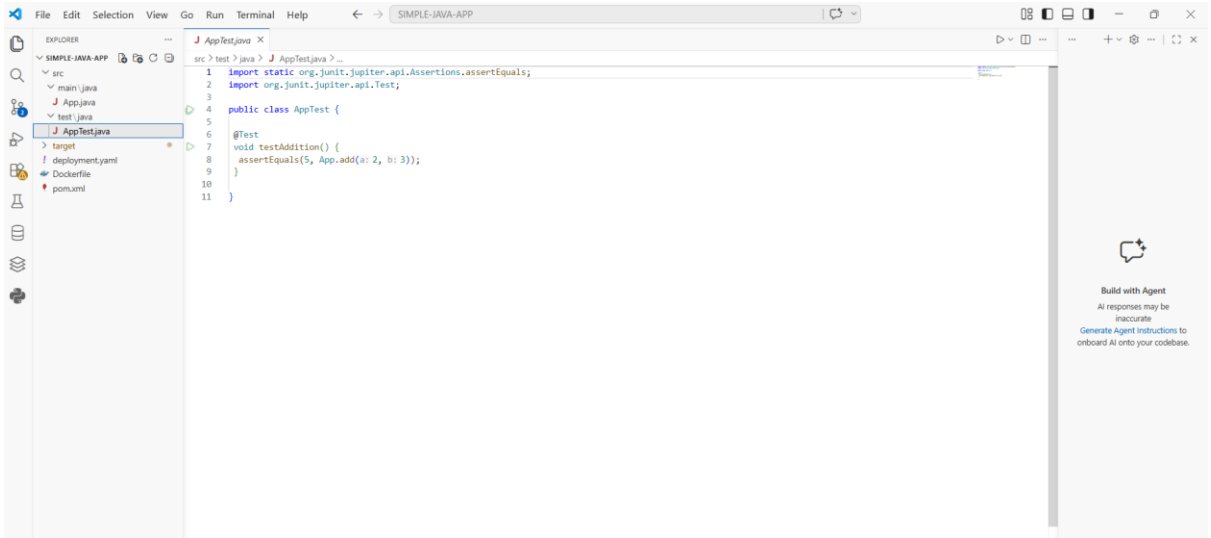

localhost:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates 14
Available plugins
Installed plugins
Advanced settings
Download progress
Preparation
Checking internet connectivity
Checking update center connectivity
Success
Authentication Tokens API Success
Docker Commons Success
Docker Pipeline Success
Cloud Statistics Success
JSON Path API Success
Token Macro Success
EDDSA API Success
Trilead API Success
SSH Build Agents Success
Apache HttpComponents Client 5.x API Success
Commons Compress API Success
Docker API Success
Docker Success
Loading plugin extensions Success
OkHttp Success
Kubernetes Client API Success
Kubernetes Credentials Success
Kubernetes CLI Success
Loading plugin extensions Success

Step 2 - Create Simple Java Application (in local machine)

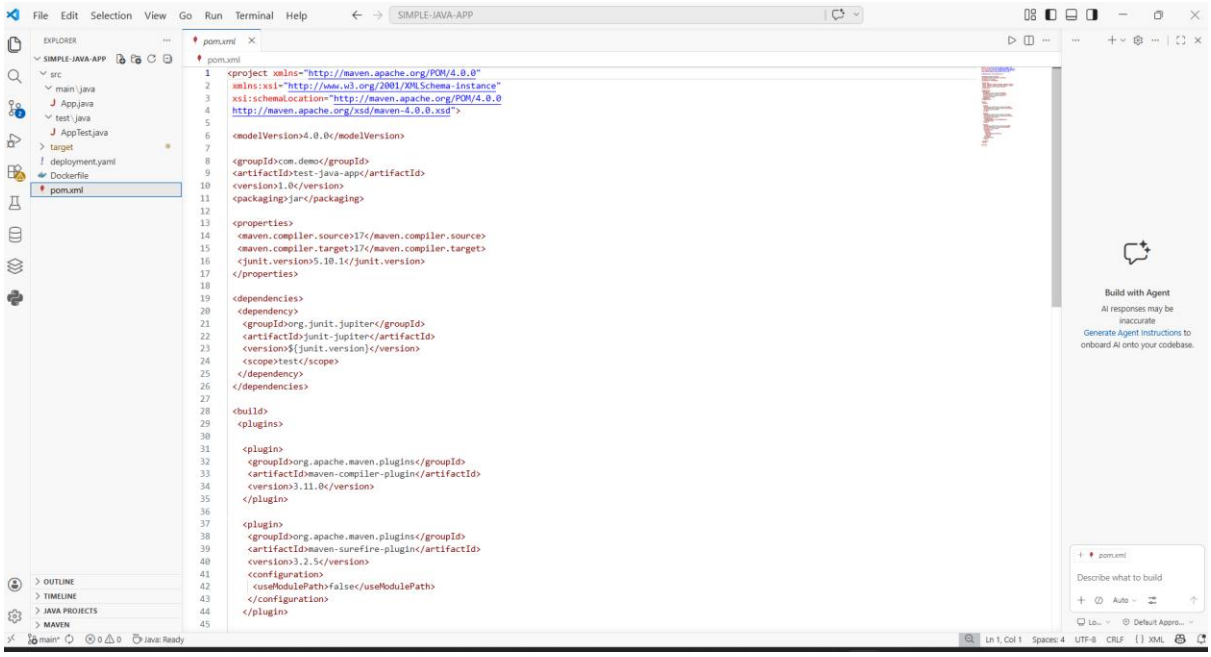
Create App.java



Create AppTest.java



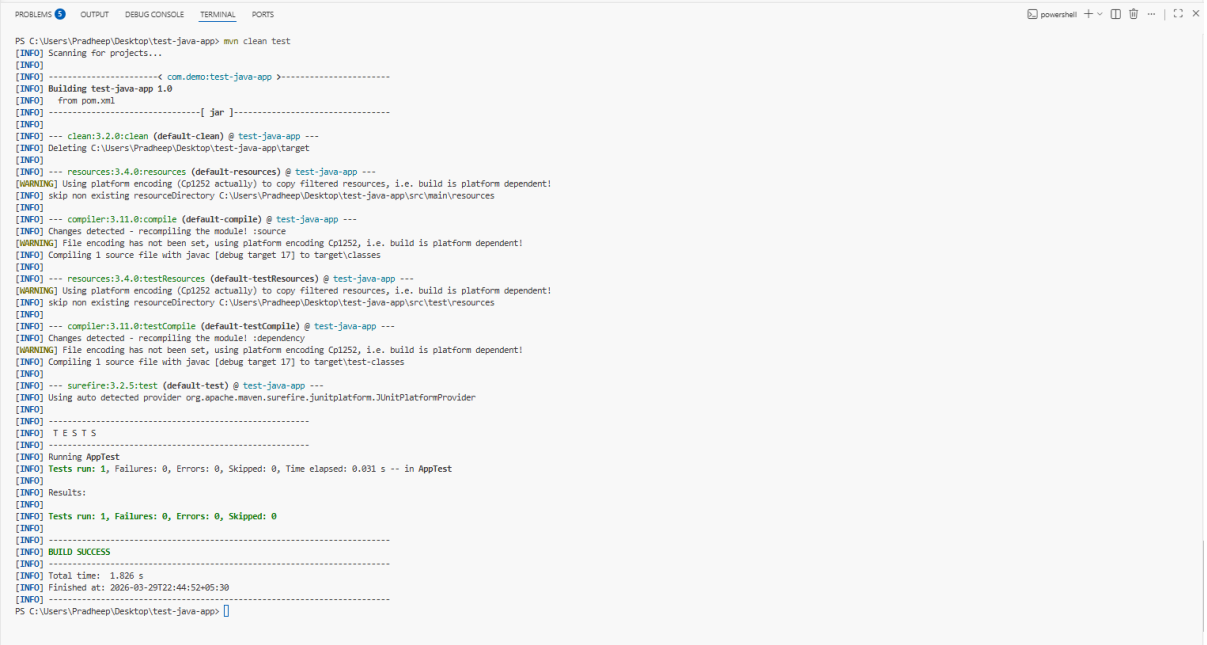
Create pom.xml



Step 3 - Build Locally using Maven (Test)

Inside project folder:

mvn clean test:



```
PROBLEMS 5   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS C:\Users\Pradheep\Desktop\test-java-app> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\Pradheep\Desktop\test-java-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\test-java-app\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\test-java-app\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.031 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.826 s
[INFO] Finished at: 2026-03-29T22:44:52+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\test-java-app>
```

mvn clean package

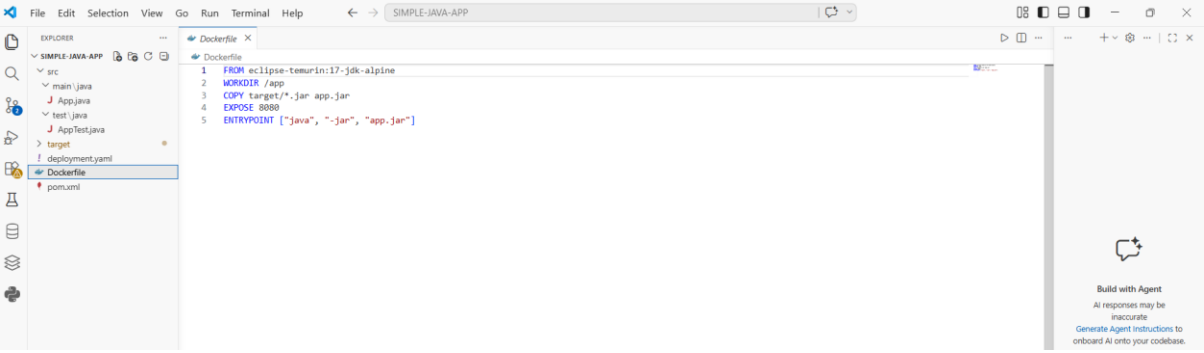
```
PS C:\Users\Pradheep\Desktop\test-java-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\Pradheep\Desktop\test-java-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\test-java-app\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\test-java-app\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.032 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\Pradheep\Desktop\test-java-app\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.147 s
[INFO] Finished at: 2026-03-29T22:54:05+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\test-java-app>
```

It generates: target/ test-java-app-1.0.jar

Name	Date modified	Type	Size
src	29-03-2026 22:03	File folder	
target	29-03-2026 22:54	File folder	
pom.xml	29-03-2026 22:43	Microsoft Edge HT...	2 KB

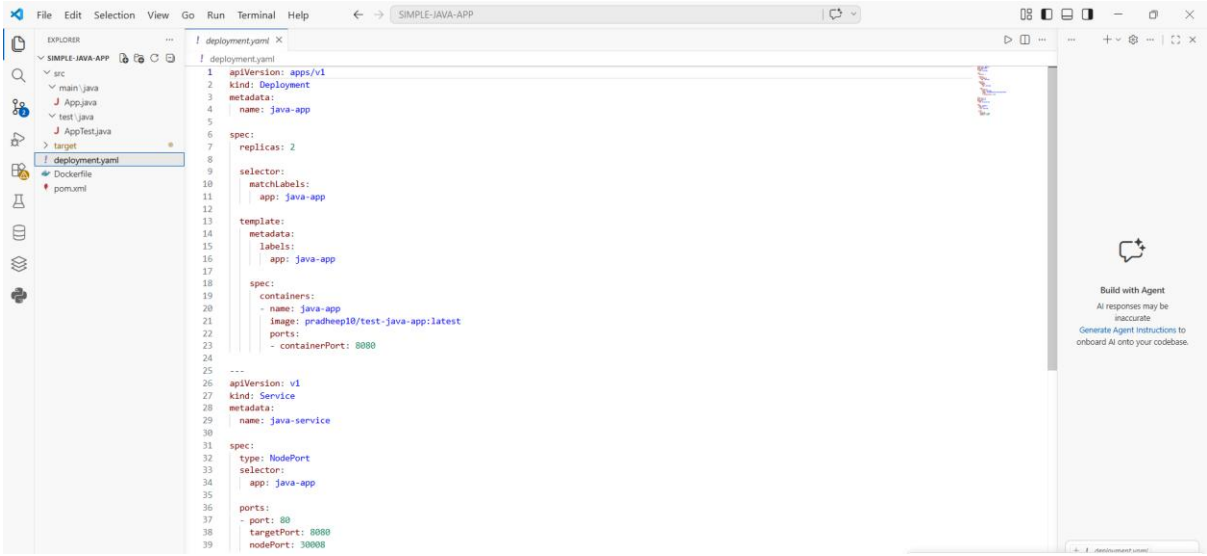
Name	Date modified	Type	Size
classes	29-03-2026 22:54	File folder	
generated-sources	29-03-2026 22:54	File folder	
generated-test-sources	29-03-2026 22:54	File folder	
maven-archiver	29-03-2026 22:54	File folder	
maven-status	29-03-2026 22:54	File folder	
surefire-reports	29-03-2026 22:54	File folder	
test-classes	29-03-2026 22:54	File folder	
test-java-app-1.0.jar	29-03-2026 22:54	Executable Jar File	3 KB

Step 4 - Create Dockerfile

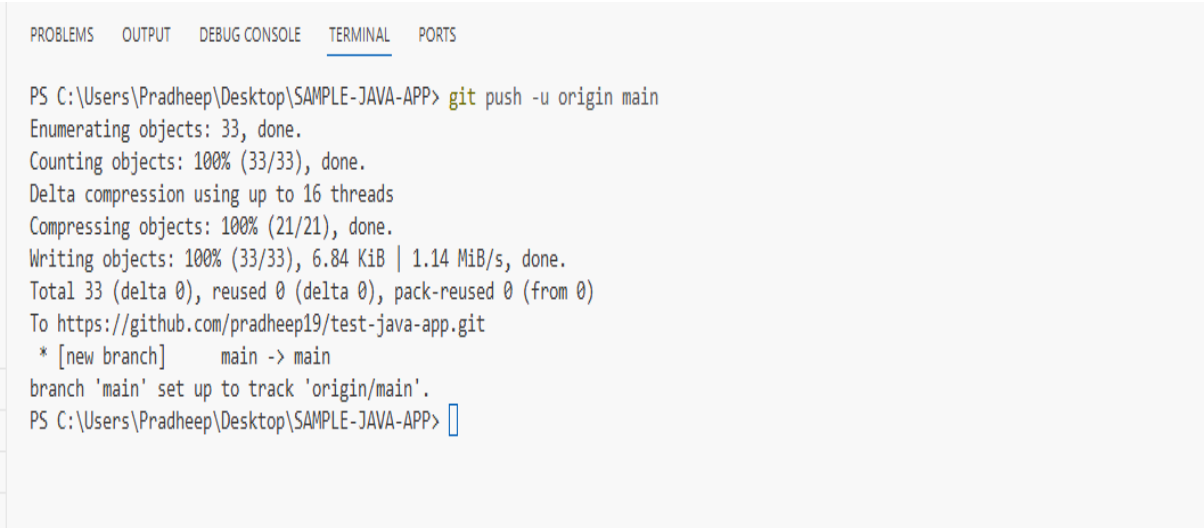


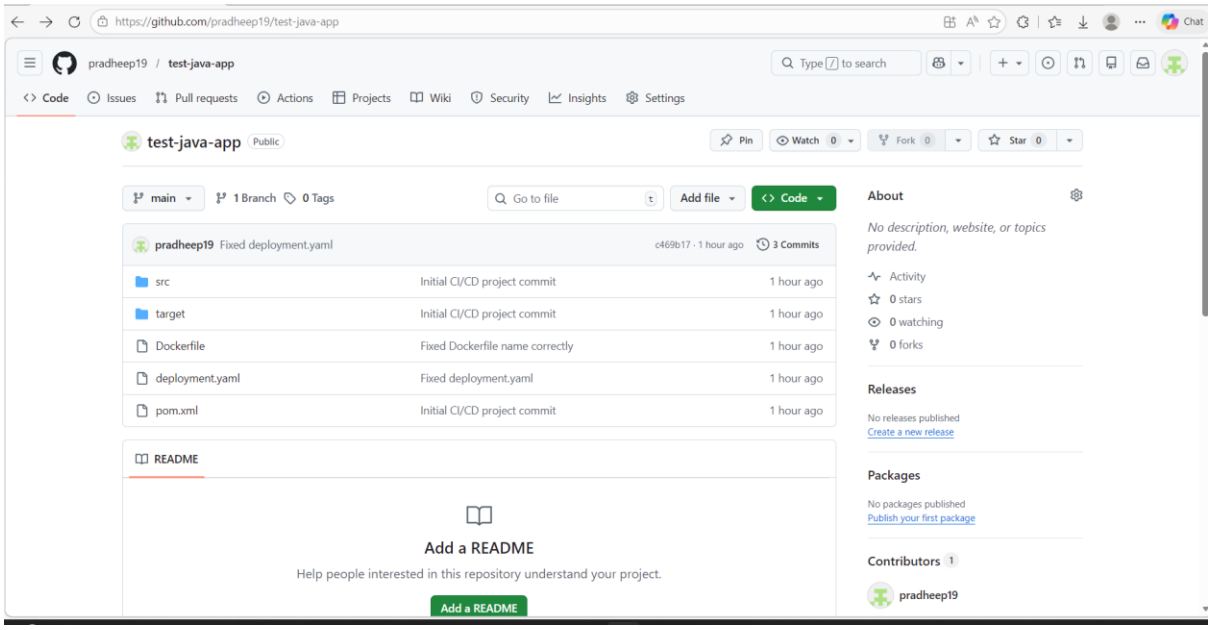
Step 5 - Create Kubernetes Deployment File

deployment.yaml

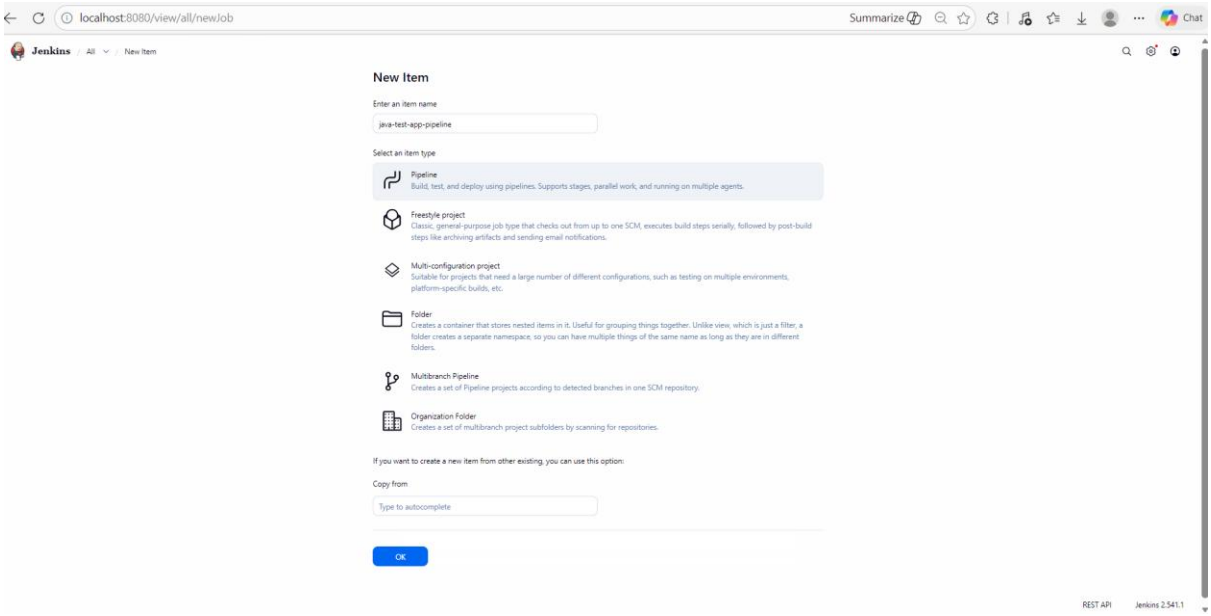


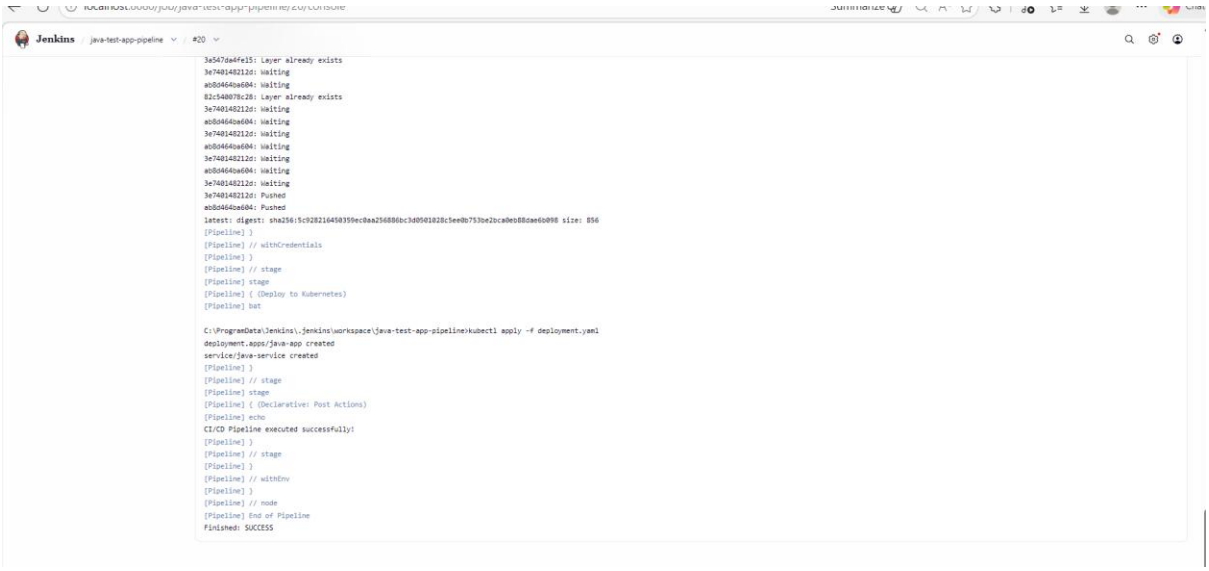
Step 6 - Push Code to GitHub

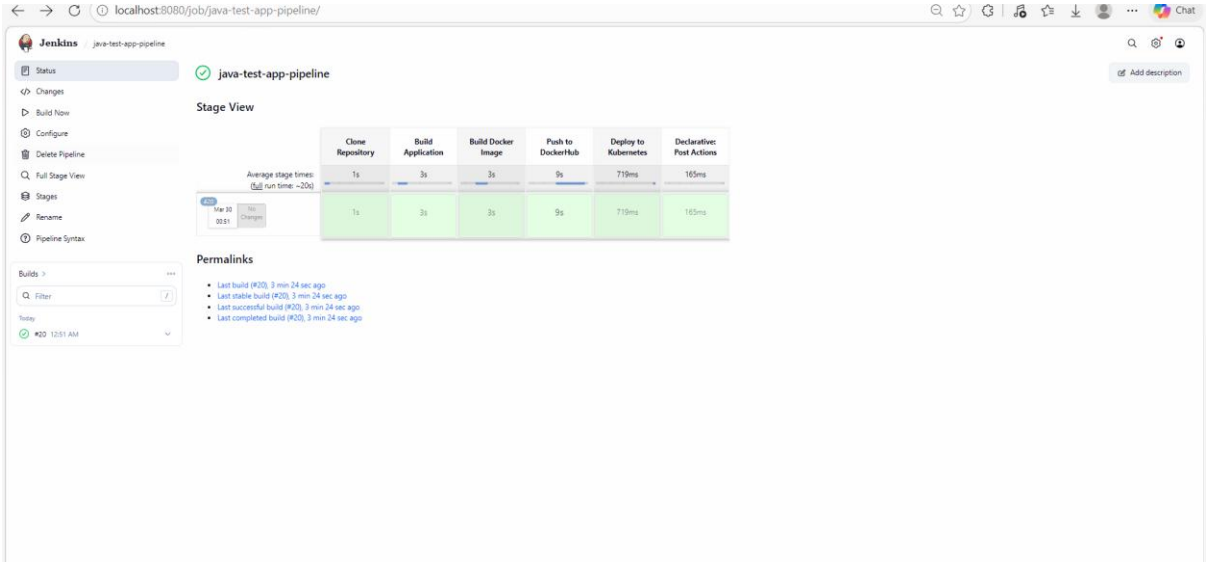




Step 7 - Create Jenkins Pipeline Job

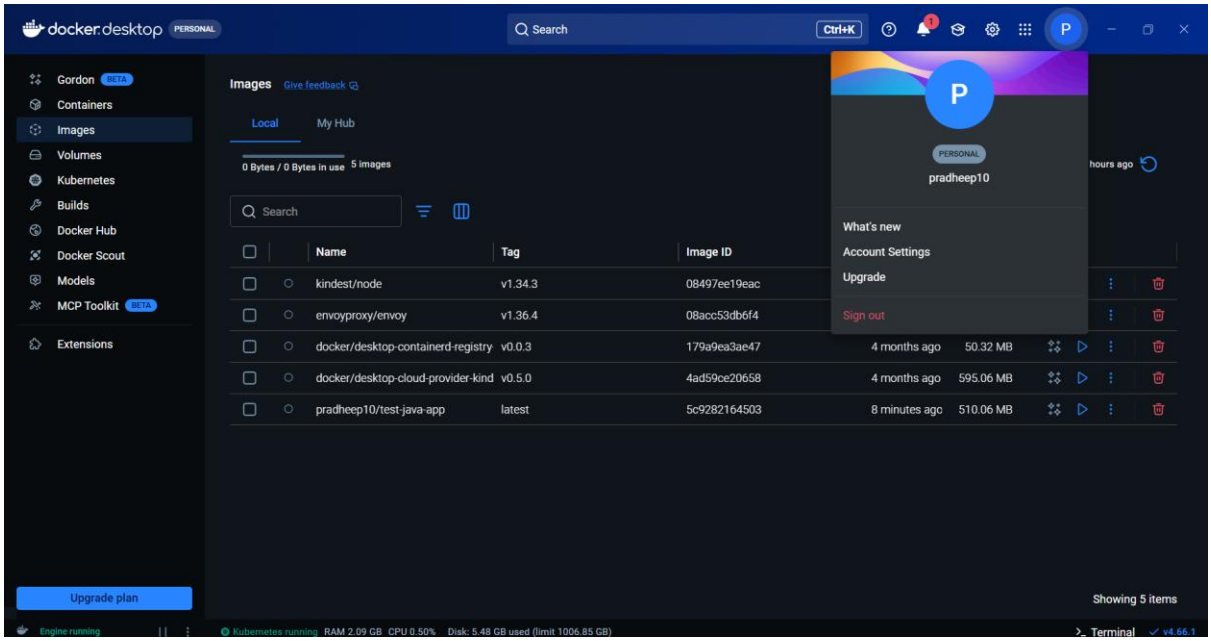




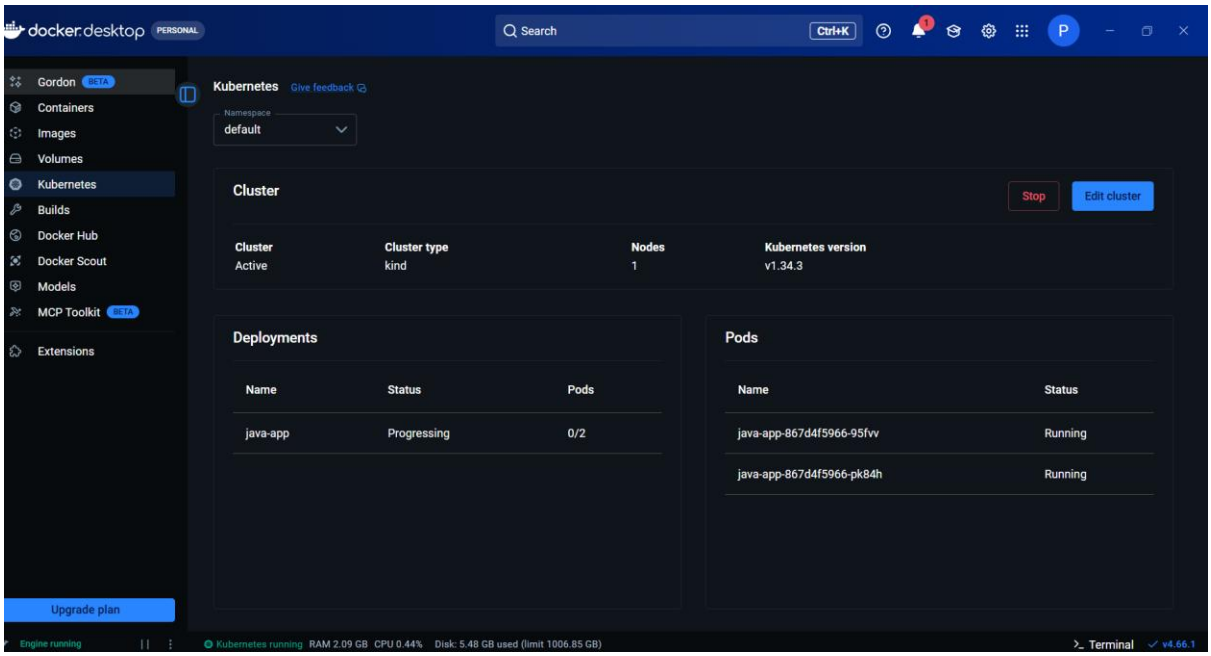


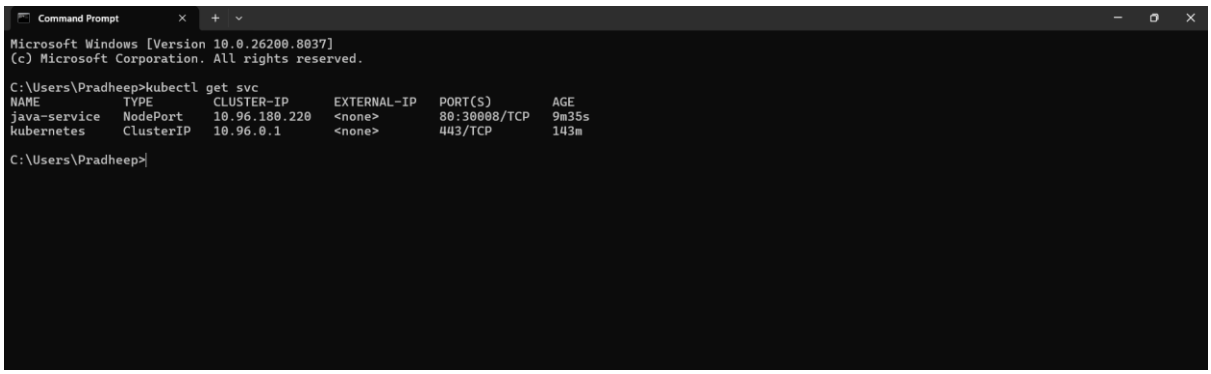
Step 12 - Verify Docker and Kubernetes Deployment

Docker Image


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Images Give feedback
Local
My Hub
0 Bytes / 0 Bytes in use 5 images
Search
Name
Tag
Image ID
kindest/node v1.34.3 08497ee19eac
envoyproxy/envoy v1.36.4 08acc53db6f4
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
docker/desktop-cloud-provider-kind v0.5.0 4ad59ce20658 4 months ago 595.06 MB
pradheep10/test-java-app latest 5c9282164503 8 minutes ago 510.06 MB
PERSONAL
pradheep10
What's new
Account Settings
Upgrade
Sign out
hours ago
Showing 5 items
Engine running
Kubernetes running RAM 2.09 GB CPU 0.50% Disk: 5.48 GB used (limit 1006.85 GB)
Terminal
v4.66.1

Kubernetes


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Kubernetes Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.34.3
Deployments
Name
Status
Pods
java-app
Progressing
0/2
Pods
Name
Status
java-app-867d4f5966-95fvv
Running
java-app-867d4f5966-pk84h
Running
Engine running
Kubernetes running RAM 2.09 GB CPU 0.44% Disk: 5.48 GB used (limit 1006.85 GB)
Terminal
v4.66.1


Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Pradheep>kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
java-service NodePort 10.96.180.220 <none> 80:30008/TCP 9m35s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 143m
C:\Users\Pradheep>